Symmetry d-matrices: orthogonal, augmented, non-orthogonal

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Setup

Symmetry acts on physical states; the relation is between all-electron (AE) orbitals. For a point-group operation $\hat{R}$ mapping IBZ point k_1 to full-BZ point $k_2 = R\,k_1$,

$$\psi_n^{k_2,\mathrm{AE}}(\mathbf{r}) \;=\; \sum_m D_{mn}(R,k_1)\,(\hat{R}\,\psi_m^{k_1,\mathrm{AE}})(\mathbf{r}), \qquad (\hat{R}\,\psi_m^{k_1,\mathrm{AE}})(\mathbf{r}) = \psi_m^{k_1,\mathrm{AE}}(R^{-1}\mathbf{r}). \tag{1}$$

Define

$$F_{pn}(R,k_1) \;:=\; \langle \hat{R}\,\psi_p^{k_1,\mathrm{AE}} | |\psi_n^{k_2,\mathrm{AE}}\rangle_{\mathrm{AE}}, \tag{2}$$

$$M_{pm}(R,k_1) \;:=\; \langle \hat{R}\,\psi_p^{k_1,\mathrm{AE}} | |\hat{R}\,\psi_m^{k_1,\mathrm{AE}}\rangle_{\mathrm{AE}}, \tag{3}$$

where $\langle\cdot|\cdot\rangle_{\mathrm{AE}}$ is the inner product on the all-electron Hilbert space. Combining with Eq. (**??**),

$$F \;=\; M\,D, \qquad D \;=\; M^{-1}\,F. \tag{4}$$

In NCPP the AE orbitals coincide with the pseudo ones, $\psi_n^{k,\mathrm{AE}} \equiv \widetilde{\psi}_n^k$, and the AE inner product reduces to the plane-wave dot product on the smooth coefficients. In USPP/PAW the link between the AE and the pseudo orbital is the projector-augmentation transformation $|\psi_n^{k,\mathrm{AE}}\rangle = \widehat{T}\,|\widetilde{\psi}_n^k\rangle$, with

$$\widehat{T} \;=\; 1 \;+\; \sum_{a,L}\big(|\phi_L^{\mathrm{AE},a}\rangle - |\phi_L^{\mathrm{PS},a}\rangle\big)\langle p_L^a|, \tag{5}$$

so that the AE inner product becomes the augmented pseudo inner product

$$\langle \psi_a^{k,\mathrm{AE}} | |\psi_b^{k,\mathrm{AE}}\rangle_{\mathrm{AE}} \;=\; \langle \widetilde{\psi}_a^k | S^{(k)} | \widetilde{\psi}_b^k\rangle, \qquad S^{(k)} \;=\; \widehat{T}^\dagger \widehat{T} \;=\; 1 + \sum_{a,L,L'} |p_L^{a,k}\rangle\, Q_{LL'}^a\,\langle p_{L'}^{a,k}|, \tag{6}$$

with $Q_{LL'}^a = \langle \phi_L^{\mathrm{AE},a} | |\phi_{L'}^{\mathrm{AE},a}\rangle - \langle \phi_L^{\mathrm{PS},a} | |\phi_{L'}^{\mathrm{PS},a}\rangle$.

Case 1 — NCPP, orthonormal AE bands

$\widehat{T} = 1$, $\psi_n^{k,\mathrm{AE}} \equiv \widetilde{\psi}_n^k$, $\langle \psi_p^{k,\mathrm{AE}} | |\psi_q^{k,\mathrm{AE}}\rangle_{\mathrm{AE}} = \delta_{pq}$. Then $M = \mathbb{I}$ and

$$\boxed{D_{pn}(R,k_1) \;=\; \sum_{\mathbf{G}} \big[\hat{R}\,\widetilde{\psi}_p^{k_1}(\mathbf{G})\big]^*\,\widetilde{\psi}_n^{k_2}(\mathbf{G}).} \tag{7}$$

Case 2 — USPP/PAW, AE-orthonormal bands

The QE-shipped bands satisfy $\langle \psi_p^{k,\mathrm{AE}} || \psi_q^{k,\mathrm{AE}} \rangle_{\mathrm{AE}} = \langle \widetilde{\psi}_p^{k} | S^{(k)} | \widetilde{\psi}_q^{k} \rangle = \delta_{pq}$, so $M = \mathbb{I}$ and

$$\boxed{\begin{aligned} D_{pn}(R,k_1) \;&= \langle \hat{R}\,\psi_p^{k_1,\mathrm{AE}} || \psi_n^{k_2,\mathrm{AE}} \rangle_{\mathrm{AE}} \;=\; \langle \hat{R}\,\widetilde{\psi}_p^{k_1} | S^{(k_2)} | \widetilde{\psi}_n^{k_2} \rangle \\ &= \sum_{\mathbf{G}} \big[\hat{R}\,\widetilde{\psi}_p^{k_1}(\mathbf{G})\big]^* \,\widetilde{\psi}_n^{k_2}(\mathbf{G}) \\ &\quad + \sum_{a,L,L'} \langle \hat{R}\,\widetilde{\psi}_p^{k_1} || p_L^{a,k_2} \rangle \, Q_{LL'}^{a} \, \langle p_{L'}^{a,k_2} || \widetilde{\psi}_n^{k_2} \rangle. \end{aligned}} \tag{8}$$

Case 3 — non-orthogonal AE bands

Drop the AE-orthonormality assumption; keep both F and M in the master equation (**??**):

$$\boxed{D(R,k_1) \;=\; \big[M(R,k_1)\big]^{-1}\, F(R,k_1).} \tag{9}$$

Written in the pseudo basis (covers NCPP, USPP, and PAW simultaneously),

$$\begin{aligned} F_{pn} \;&=\; \langle \hat{R}\,\widetilde{\psi}_p^{k_1} | S^{(k_2)} | \widetilde{\psi}_n^{k_2} \rangle \\ &=\; \sum_{\mathbf{G}} \big[\hat{R}\,\widetilde{\psi}_p^{k_1}(\mathbf{G})\big]^* \widetilde{\psi}_n^{k_2}(\mathbf{G}) \;+\; \sum_{a,L,L'} \langle \hat{R}\,\widetilde{\psi}_p^{k_1} || p_L^{a,k_2} \rangle \, Q_{LL'}^{a} \, \langle p_{L'}^{a,k_2} || \widetilde{\psi}_n^{k_2} \rangle, \end{aligned} \tag{10}$$

$$\begin{aligned} M_{pm} \;&=\; \langle \hat{R}\,\widetilde{\psi}_p^{k_1} | S^{(k_2)} | \hat{R}\,\widetilde{\psi}_m^{k_1} \rangle \\ &=\; \sum_{\mathbf{G}} \big[\hat{R}\,\widetilde{\psi}_p^{k_1}(\mathbf{G})\big]^* \hat{R}\,\widetilde{\psi}_m^{k_1}(\mathbf{G}) \;+\; \sum_{a,L,L'} \langle \hat{R}\,\widetilde{\psi}_p^{k_1} || p_L^{a,k_2} \rangle \, Q_{LL'}^{a} \, \langle p_{L'}^{a,k_2} || \hat{R}\,\widetilde{\psi}_m^{k_1} \rangle. \end{aligned} \tag{11}$$

For NCPP the augmentation terms vanish ($Q \equiv 0$); the equations reduce to the smooth dot products of the pseudo (= AE) coefficients.

Evaluating the augmentation term

Should this involve $\widehat{T}_{k_1}^{\dagger}\,\widehat{T}_{k_2}$? No — $\widehat{T}$ is a single, real-space, k-independent operator, $\widehat{T} = 1 + \sum_{a,L}(|\phi_L^{\mathrm{AE},a}\rangle - |\phi_L^{\mathrm{PS},a}\rangle)\langle p_L^a|$, where the sums run over all atoms in the crystal (every periodic image) and the partial waves/projectors are the same atom-localised functions used for every k. The apparent k-dependence enters only through Bloch summation over the periodic images when one projects a Bloch state against those atom-local projectors.

In our case both states in $F_{pn} = \langle \hat{R}\,\widetilde{\psi}_p^{k_1} | \widehat{T}^{\dagger}\widehat{T} | \widetilde{\psi}_n^{k_2} \rangle$ carry crystal momentum k_2: $\hat{R}\,\widetilde{\psi}_p^{k_1}$ has crystal momentum $R\,k_1 = k_2$ by construction, and $\widetilde{\psi}_n^{k_2}$ is at k_2 already. Performing the sum over all atom positions $\tau_\alpha + T$ with α a unit-cell atom and T a lattice translation, the Bloch translation phases combine as $e^{-ik_2\cdot T}e^{+ik_2\cdot T} = 1$, leaving

$$F_{pn}^{\mathrm{aug}} \;=\; \sum_{\alpha,L,L'} \langle \hat{R}\,\widetilde{\psi}_p^{k_1} || p_L^{\alpha,k_2} \rangle \, Q_{LL'}^{\alpha} \, \langle p_{L'}^{\alpha,k_2} || \widetilde{\psi}_n^{k_2} \rangle, \tag{12}$$

where the sum is over unit-cell atoms α and the projectors $|p_L^{\alpha,k_2}\rangle$ are the standard Bloch-summed, unit-cell-restricted projectors at k_2:

$$|p_L^{\alpha,k}\rangle \;=\; \frac{1}{\sqrt{N_{\mathrm{cell}}}} \sum_{T} e^{ik\cdot T}\, |p_L^{\alpha}(\mathbf{r} - \boldsymbol{\tau}_\alpha - \mathbf{T})\rangle. \tag{13}$$

The right-hand projection $\langle p_{L'}^{\alpha,k_2} | | \widetilde{\psi}_n^{k_2} \rangle$ is the standard `Pskna` entry at k_2, already in storage. The new piece is the left-hand projection $\langle \hat{R}\, \widetilde{\psi}_p^{k_1} | | p_L^{\alpha,k_2} \rangle$, between an orbital indexed by k_1 (its storage label) and a projector at k_2 (its crystal-momentum label after rotation).

Why $k_1 \neq k_2$ does not break the calculation. If we did the same exercise with $\langle \widetilde{\psi}_p^{k_1} | S | \widetilde{\psi}_n^{k_2} \rangle$ *without* the rotation, the Bloch-translation phase sum gives $\sum_T e^{i(k_2-k_1)\cdot T} = N_{\text{cell}}\, \delta_{k_1,k_2 \bmod \mathbf{G}}$, which vanishes for distinct k_1, k_2 as it must for an orthonormal Bloch basis. The rotation $\hat{R}$ shifts the bra-side crystal momentum from k_1 to k_2, restoring the matching that makes the overlap non-trivial.

Strategy A — direct re-projection at k_2. Apply the same $\mathbf{G}$-vector rotation that builds $\hat{R}\, \widetilde{\psi}_p^{k_1}(\mathbf{G})$ for the smooth term (already in the code at `transform_k2g`), and reuse the standard projector machinery at k_2 to compute

$$\langle \hat{R}\, \widetilde{\psi}_p^{k_1} | | p_L^{\alpha,k_2} \rangle \;=\; \sum_{\mathbf{G}} \big[\hat{R}\, \widetilde{\psi}_p^{k_1}(\mathbf{G}) \big]^* \, p_L^{\alpha,k_2}(\mathbf{G})\, e^{-i(\mathbf{G}+k_2)\cdot \boldsymbol{\tau}_\alpha}. \tag{14}$$

Pros: reuses the FFT-rotated coefficients already needed for the smooth term; no symmetry-property machinery for the projectors. Cons: costs an additional projector sweep per (R, k_1) pair.

Strategy B — symmetry-routed projection at k_1. Use the transformation property of the projector under $\hat{R}$. With R acting on atom α as $\hat{R}\,(\boldsymbol{\tau}_\alpha) = \boldsymbol{\tau}_{R\alpha} + \mathbf{t}_R^\alpha$ (the residual translation $\mathbf{t}_R^\alpha$ takes the rotated atom into the same unit cell), and $D_{ML}^{(l)}(R)$ the Wigner D-matrix of the angular momentum $l = l_L$:

$$\hat{R}\, | p_L^{\alpha,k_1} \rangle \;=\; e^{-ik_2\cdot \mathbf{t}_R^\alpha} \sum_M D_{ML}^{(l)}(R)\, | p_M^{R\alpha,\,k_2} \rangle. \tag{15}$$

Hence

$$\langle \hat{R}\, \widetilde{\psi}_p^{k_1} | | p_L^{\alpha,k_2} \rangle \;=\; \langle \widetilde{\psi}_p^{k_1} | \hat{R}^\dagger | p_L^{\alpha,k_2} \rangle \;=\; e^{+ik_2\cdot \mathbf{t}_R^{R^{-1}\alpha}} \sum_M \big[D_{ML}^{(l)}(R) \big]^* \, \langle \widetilde{\psi}_p^{k_1} | | p_M^{R^{-1}\alpha,\,k_1} \rangle. \tag{16}$$

Pros: the projector overlaps on the right of Eq. (**??**) are exactly the standard `Pskna` entries at k_1, already cached — no FFT rotation of the wavefunction, no recomputation of projector convolutions. Cost is only the per-(R, α, L) Wigner rotation. Cons: requires the lookup tables $R\alpha$, $\mathbf{t}_R^\alpha$, $D^{(l)}(R)$ to be in place.

Strategy C — precompute the rotated `Pskna` table. A practical hybrid: Strategy B is applied once per (R, k_1) pair to build a rotated projector-overlap table

$$\widetilde{P}_{p,L,\alpha}^{(R,k_1)} \;:=\; \langle \hat{R}\, \widetilde{\psi}_p^{k_1} | | p_L^{\alpha,k_2} \rangle \tag{17}$$

which is then reused for all n in Eq. (**??**). Cost: same as Strategy B for the table build, then a single `GEMM` per (R, k_1) block to contract with $Q_{LL'}^\alpha$ and the k_2-side $\langle p_{L'}^{\alpha,k_2} | | \widetilde{\psi}_n^{k_2} \rangle$. This is the recommended layout: it keeps the full D-matrix as a single GEMM plus the augmentation correction as a second GEMM of the same shape.

Recap of what is needed. Implementation footprint for the QE-source PAW workflow ($M = \mathbb{I}$):

- The species-resolved $Q^{\alpha}_{LL'}$ is already loaded in `pseudopot::qq_nt_data`.
- The right-hand $\langle p^{\alpha,k_2}_{L'} || \widetilde{\psi}^{k_2}_{n} \rangle$ is the standard `Pskna` entry at k_2 (already cached).
- The left-hand $\langle \hat{R}\, \widetilde{\psi}^{k_1}_{p} || p^{\alpha,k_2}_{L} \rangle$ is the new piece. Strategy A and C share the same final contraction; they differ only in how the table is built (FFT rotation vs. Wigner-D symmetry routing). Strategy B/C is cheaper if the symmetry tables are already on hand.